DROP
Filtration systems
Irrigation equipment

SELF CLEANING
SCREEN FILTERS
SUFV



VERSION 1.4

DROP
Filtration systems
Irrigation equipment

INDEX

INTRODUCTION 4

RECOMMENDATIONS 5

SPECIFICATIONS 6

SAFETY PRECAUTIONS 6

DIMENSIONS 7

OPERATING PRINCIPLE 8

PUT INTO OPERATION 9

CONTROLLER INSTRUCTIONS 11

- GENERAL 11
- SETTINGS 12
- MANUAL CLEANING 16

MAINTENANCE 16

WIRING DIAGRAM 19

- DC MODEL 19
- AC MODEL 20

HYDRAULIC CONNECTIONS 21

Lloyd's Register

Certificate of Approval

This is to certify that the Management System of:

DROP - N. KIOSIDIS S.A.

Industrial Area of Thessaloniki, Sindos, 57022 Thessaloniki, Greece

has been approved by LRQA to the following standards:

BS EN ISO 9001:2015



Gilles Bessiere - Area Technical Manager

Issued By: Hellenic Lloyd's S.A.

for and on behalf of: Lloyd's Register Quality Assurance Limited

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The scope of this approval is applicable to:
Design and production of steel filters for irrigation and water supply use. Sales of irrigation equipment.

UKAS MANAGEMENT SYSTEMS
001

INTRODUCTION

The company DROP -N.KIOSIDIS SA thanks you for purchasing this product.

All our company's products are manufactured with the most appropriate materials in accordance with strict standards and high quality specifications.

The production process and the quality of the company's products are controlled in accordance with the Quality Management System ISO 9001, which is certified by Lloyds.

Our quality products, productive capabilities, strong organization and the qualified technical personnel, provide us a dynamic and optimistic future.

The combination of the production of high quality products along with high corporate objectives and dynamic capable personnel, have made our company's name an International and instantly recognizable symbol.

For the proper installation and operation of the system and to avoid any problems it is necessary to study carefully this manual before installing the system

This manual is an inseparable and essential part of the system and provides you instructions and important information regarding the safe installation, operation, and maintenance. It should be kept in a safe place along with the system so at any time to be able to consult the user.

Any suggestion from you about our products or the handbook is welcomed.

Our Company and our specialized technical personnel are at your disposal for any additional information you may need about our products

RECOMMENDATIONS

- In case of fluctuating flow or varying downstream pressure, it is advisable to install a pressure sustaining valve downstream the filter in order to keep the minimum backpressure required for efficient cleaning.
- Where the filter operates in low pressure installations, a main valve that will be partially or totally closed should be installed downstream the filter. It should operate after a command of filtration controller during cleaning process.
- For safety reasons or when in extreme cases the filter can be totally clogged, install a quick pressure relief valve upstream the filter to avoid overpressure. The valve should be adjusted to open at one (1) bar over the normal working pressure or maximum acceptable pressure of filter
- In order to avoid trapped air in the piping system, it is advisable to install air release valve(s) in certain points of the network

To protect from reverse flow, install a check valve (preferably membrane type) downstream the filter.

SPECIFICATIONS

CODE	DESCRIPTION	CONNECTIONS	FILTRATION AREA	FLOW	CLEANING FLOW RATE	MAX PRESSURE	MIN PRESSURE
			cm^2	m^3/h	m^3/h	bar	bar
05-11-201	Self cleaning screen filter vertical series SUFV 2"	TF	780	25	15	10	2
05-11-301	Self cleaning screen filter vertical series SUFV 3"	F	1450	40	20	10	2
05-11-401	Self cleaning screen filter vertical series SUFV 4"	F	1450	80	20	10	2

MATERIALS

Body: Steel S235JR (also available in stainless steel AISI 304 & 316, on request)

Cartridge: Perforated stainless steel sheet AISI 304, Molded Weave Wire stainless steel AISI 316 100 micron, rigid pvc (u-pvc)-pvc100 (DIN8061-8062)

Gaskets: EPDM

Surface treatment: Chemical cleaning (pickling, phosphate) and electrostatic painting with polyester. powder (available epoxy coatings suitable for drinking water).

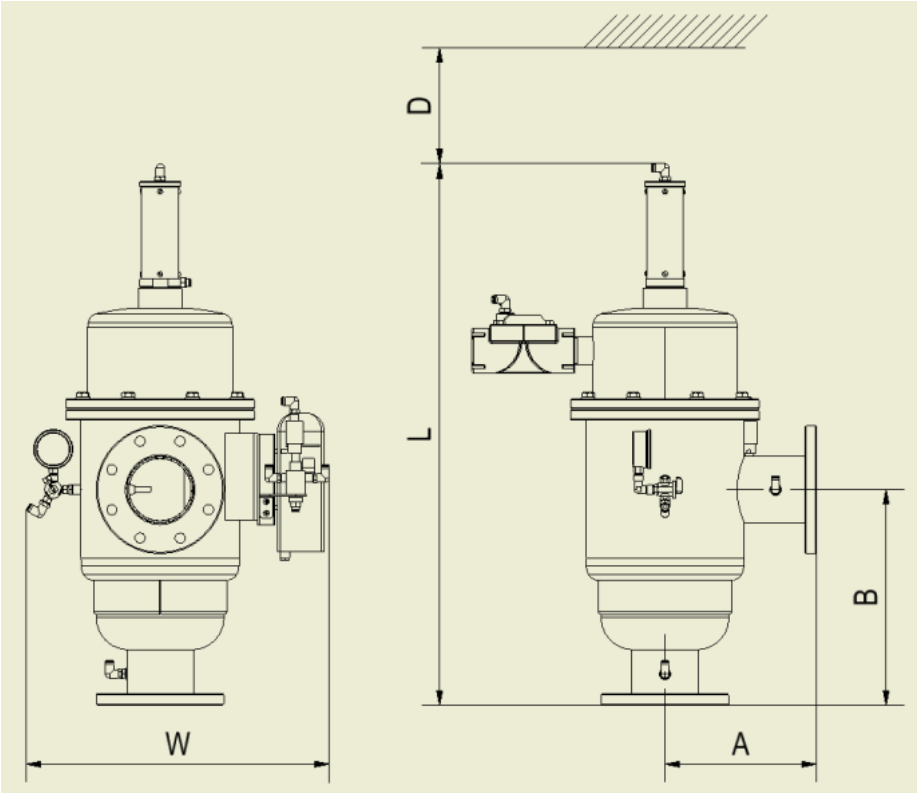
SAFETY PRECAUTIONS

When installing or during operation of the system, the following safety precautions should be followed:

- Properly read and understand these instructions before installation and operation
- Take all necessary safety measures depending on the environment and the equipment used (eg distance from electric fields, forklifts, etc.)
- While the filter is on, the cleaning process can be initiated without prior notice
- Disassembly should always be done while the filter is off and if there is no hydrostatic pressure inside

Changes and modifications to the system are not permitted without the prior written consent of the manufacturer or his representative.

DIMENSIONS



TYPE	L	A	B	W	D	WEIGHT
	mm	mm	mm	mm	mm	Kg
Self cleaning screen filter vertical series SUFV 2"	795	200	280	510	-	48
Self cleaning screen filter vertical series SUFV 3"	935	260	370	510	-	58
Self cleaning screen filter vertical series SUFV 4"	950	260	370	510	-	60

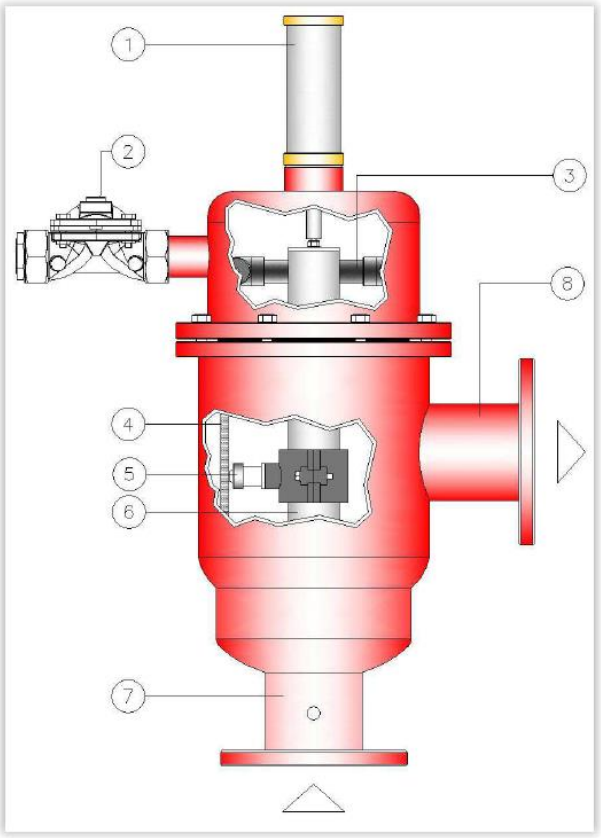
*D: Required minimum length for opening

OPERATING PRINCIPLE

FILTRATION

The water enters the filter from the bottom (7). The cartridge with the stainless steel wire mesh (4) filters the dirty water and clean water exits from the side outlet (8). During the filtration, there is pressure on the piston (1) and the chamber of the hydraulic valve (2) having as a result the axis of the nozzles (6) being in the lower position and the valve is closed.

CLEANING

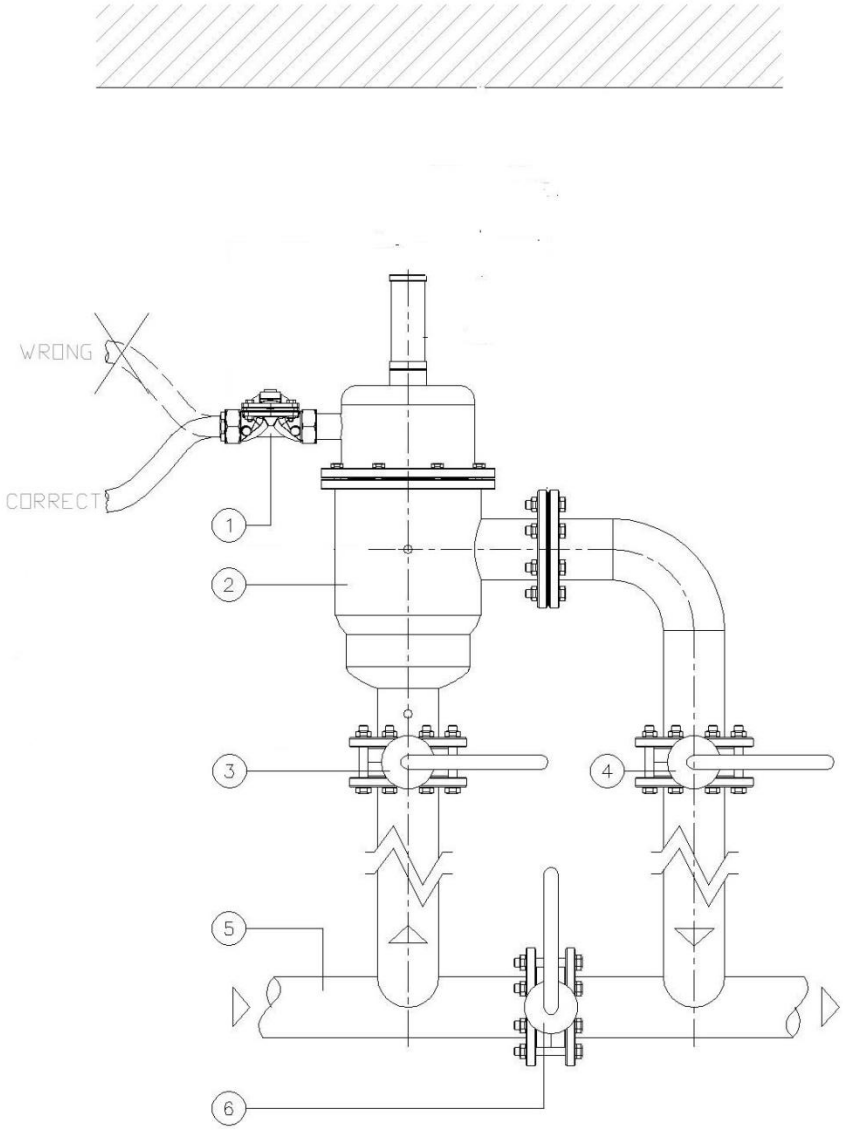


When the input-output differential pressure reaches the predetermined set point (max. 0.5 bar) and/or after the passing of the predetermined time that has been introduced to the controller, the cleaning process starts.

The pressure in the piston (1) and the valve chamber (2) is vented to the atmosphere and as a result the axis of the nozzles (6) begins to move upwards and the valve opens.

The pressure difference between the outer and inner surface of the cartridge (4), creates locally in the area at the front of the nozzle (5) high speed flow conditions which results that solids that were retained on the surface of the mesh to be entrained and driven through the axis (6) and valve (2) into drains. The flow of water into the shaft (6) and through the circumferentially nozzles (3) creates a rotational movement simultaneously to the upward movement. Thus the nozzles (5) sweep and clean the entire inner surface of the cartridge (4). The cleaning process takes 15 seconds maximum. During the process of cleaning, the water flow to the outlet is not interrupted.

PUT INTO OPERATION



1 - Drain
2 - Filter
3 – Inlet valve
4 – Outlet valve
5 – Main line
6 – By-pass valve

For safe installation read thoroughly the instructions in this manual. The necessary connections (hydraulic, electric) must be performed by qualified personnel.

1. Install the filter at horizontal position on the main pipe. One proposed method is illustrated in the above image (this method allows filter maintenance without interrupting the water supply)
2. Make sure there is sufficient space for maintenance and disassembly of the filter (see table DIMENSIONS in previous page)

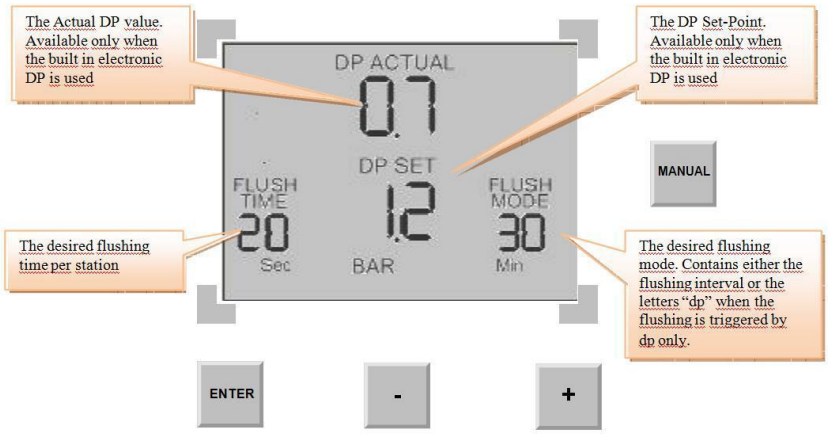
3. Make sure that the input-output connections, drainage valve are made correctly and the flow direction is parallel to the arrows that are in the filter.
4. Connect the drainage valve with a suitable recipient. The length of the sewer pipe must not exceed 10 m and not be higher than the filter
5. Do the hydraulic connections according to HYDRAULIC INSTALLATION DRAWING
6. Connect the power supply to filter controller (for AC powered models)
7. Partially open the inlet and outlet valve and watch for leaks.
8. Let the pressure rise at it's current value and finally open full both valves
9. Perform a manual cleaning by pressing the manual key in the controller (see CONTROLLER INSTRUCTIONS) and check if water flows to the drain

CONTROLLER INSTRUCTIONS



GENERAL

- The "CONTROLLER 1-10" is a modular backflushing controller for automatic filters of 1 to 10 stations.
- There exist DC and AC models.
- The DC model can be powered either by 6v DC or 12v DC and it activates 2 wired 12v DC latching solenoids.
- The AC model contains an internal transformer that can be powered by 110v or 220v from which it generates the 24v AC for the solenoids.
- Flushing cycles may be triggered either by time or by the embedded electronic DP sensor reaching the set point, or by a dry contact signal from an external DP sensor.
- Endless looping problems can be eliminated by detecting repeated consecutive cycles passing beyond a predefined limit.
- The unit can optionally handle a Pressure-Sustaining / Main valve, and an Alarm output. The unit is equipped with a customized LCD display and key board.
- The unit counts separately the number of flushing cycles triggered by DP, by time and manually.



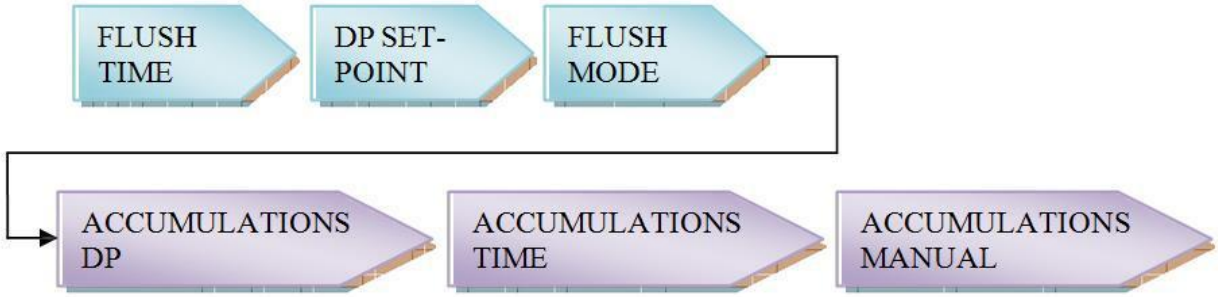
The screen consists of several fields, some of them are editable and some of them are not. For inserting EDIT MODE the ENTER key has to be pushed. The EDIT MODE is indicated by blinking of the characters at the currently editable field. Each time the ENTER key is pushed again, the next editable field becomes under focus and starts blinking. While in EDIT MODE the "+" and "-" keys can be used for changing the value under focus. Pushing the ENTER key again will set the selected value to the current field and move

the focus to the next editable field which will start blinking. Once entering this process of passing through the editable fields, the user has no way back but by pushing the ENTER key repeatedly, he passes through the chain of editable fields until arriving back to the FLUSH TIME field, meeting no more blinking fields

NOTE : Notice that before the first use of the unit, it may be necessary to pass through the configuration process prior to defining the flushing program in order to djust the features of controller to the specific application. The configuration process is described below

SETTINGS

Following is the chain of editable fields. The existence of the DP SET-POINT field depends on whether the system contains a built-in electronic DP or not.



To change the current scheduling parameters must press ENTER, and will see that flashes on the screen the first field of the parameter range. Each time you press the ENTER button you go to the next field. In the field flashing, pressing the (+) or (-) changes values and registered with ENTER.

FLUSH TIME

Defines the duration of the flushing time per station. The following options are selectable: 5-20 sec in steps of 1 sec, 20-55 sec in steps of 5 sec, 1-6 min in steps of 0.5 min

TABLE 1: Cleaning duration adjustments (suggested)

Inlet pressure (bar)	Duration (sec)
2-3	15
4-5	12
>5	10

DP SET POINT

At this field the user defines the pressure difference between the filter's inlet and outlet that when reached, a flushing cycle will take place. This field is meaningless when there is no built in electronic DP sensor included, therefore the user is expected to define the DP set point to be 00, as a result the actual DP value will appear as (- -).

When the pressure is expressed in BAR the range of values is 0.1 – 2.0 BAR. When the pressure is expressed in PSI the range of values is 1- 30 PSI.

When the system does not include the built in electronic DP sensor but uses instead an external DP sensor, the flushing request signal arrives in the shape of a closed dry contact at the appropriate input terminals

FLUSH MODE

The Flush Mode defines how the flushing cycles is triggered. The selectable options are as follows:

OFF - no flushing will take place

By time – In this case the flushing cycles will be repeated in a selected interval or will be triggered by the DP signal depending on what happens first. No matter how was the flushing cycle started the interval to the next cycle will start to be measured again after each ending of a flushing sequence. The selectable intervals are the following:

5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60

minutes 2, 3, 4, 5, 6, 8, 12, 18, 24, 72, 120 hours

dp – flushing will be triggered by DP only.

NOTE : If the "+" and "-" keys are pressed and held down simultaneously the Flush Mode" field will show the left time until next cycle, alternately hours and minutes

ACCUMULATIONS

The unit accumulates and displays the number of flushing cycles caused by DP, by time, or manually. At each of the accumulation fields, the "+" or "-" keys may be used for clearing the accumulated value.

CONFIGURATION

In order to enter into the configuration process press and hold down the ENTER key for at least 3 seconds.

The unit will detect how many "plug-in" boards (each of 2 outputs) are used in the particular case. How will the outputs be allocated depends on the definitions made during the configuration process described below. The following rules apply:

1. Backflush valves will be allocated starting from output 1 and up.
2. The last backflush valve can be canceled and then its allocated output will be left unused.
3. Alarm output, Delay-Valve and Main-Valve when defined, will be allocated in this order, right after the last backflush valve (whether in use or not).

Example:

Assuming there are 3 "plug-in" boards, this makes 6 outputs for use. If there are no Alarm-output, no Delay-Valve and no Main-Valve all the 6 outputs will be allocated for backflush valves.
If additionally a Main-Valve is defined, the first 5 outputs will be allocated for backflush valves and output No 6 for the Main-Valve. Output No 5 (of the last backflush valve) can be canceled and left unused. If additionally a Delay-Valve is defined it will be allocated to output 5 right before the Main valve, leaving the first 4 outputs for backflush valves, and once again output No 4 (of the last backflush valve) can be canceled and left unused. If additionally an Alarm-output is defined it will be allocated before the Delay-Valve leaving only 3 of the first outputs for backflush valves. No 3 can again be canceled.

During the configuration process the following features are defined:

Main Valve (sustaining valve) - Yes/ No. When the answer is "Yes" the Pre Dwell delay between the Main Valve opening and the opening of Station No. 1 can be defined. The selectable delay steps are:
5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55 sec
1, 1.5, 2, 2.5, 3, 3.5, 4, 4.5, 5, 5.5, 6 min
Dwell time - the delay between stations – can be set to 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, or 60 sec.

DP delay - the delay during which the DP sensor reading is expected to remain stable before reaction – 5, 10, 15, 20, 25, 30, 35, 40, 45, 50, 55, 60 sec.

Looping limit - the number of consecutive flushing cycles triggered by the DP sensor before deciding that there is an endless looping problem. The options are: 1-10 or "no" which means ignoring the looping problem.

Alarm - Yes/No – allocating one output for alarm activation. Delay

Valve - Yes/No – allocating an output for Delay Valve activation.

View Outputs - this is a special mode that enables passing through the list of outputs to see how each output was allocated. Use the + key to change the "no" into "yes" and confirm by "Enter", then keep using the + key to pass through the list. At the bottom left corner the ordinal number of the output is displayed and its allocated function appears in large letters at the center of the screen. Notice that the number of possible outputs that can be used is always an even number since it results from the number of "plug in" boards (each of 2 outputs) included. However if the number of outputs needed is not an even number, then the last valve allocated for flushing may be canceled by use of the  manual operations key.

Pressure units - deciding about the units to be used for pressure measurement. Selecting between BAR or PSI .

Calibration- Zero calibration of the built in electronic DP sensor. While the sensor ports are disconnected select Calibration = Yes.

ENDLESS LOOPING PROBLEMS

As explained above, endless looping problem will be declared when the number of consecutive flushing cycles triggered by the DP sensor exceeds the "Looping limit" defined during configuration. The fact that endless looping problem was detected will be indicated on the display and will cause the activation of the Alarm output, additionally, the DP indication will no longer be considered as a trigger for flushing. The following flushing cycles will be triggered by the interval count down only.

The problem will be considered as solved when the constant indication of the DP sensor will be removed.

LOW PRESSURE

When a closed contact indication is received at the low pressure input of the controller, the symbol  will start to appear blinking at the display. All activities will stop including the countdown to the

next flushing cycle. If the low pressure happened while a flushing sequence was in progress, when the low pressure condition terminates the flushing sequence will start from the beginning rather than continue from the stop point.

LOW BATTERY

The unit has two levels of low battery indication. At the first level when the battery voltage drops to the first level, the sign  will start to appear at the screen. When the battery voltage drops further and reaches the second level, all outputs will shut down, the screen will be cleared leaving only the low battery icon.

MANUAL CLEANING

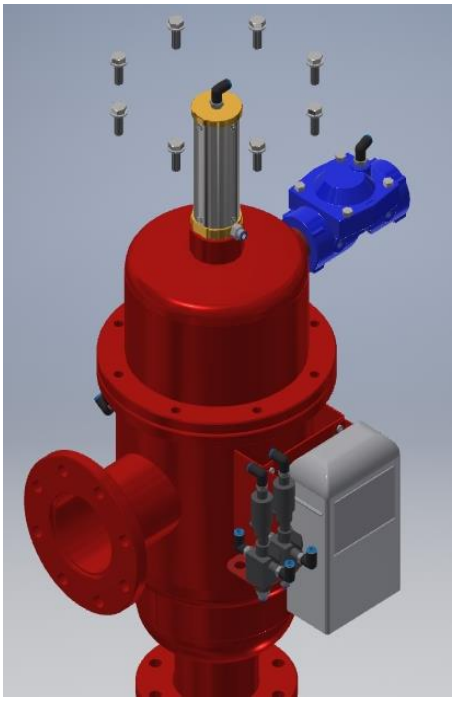
A flushing sequence can be manually activated by the "MANUAL" key. When manually activated the icon  will appear on the display. The same key will be used for manually terminating a sequence in progress.

MAINTENANCE

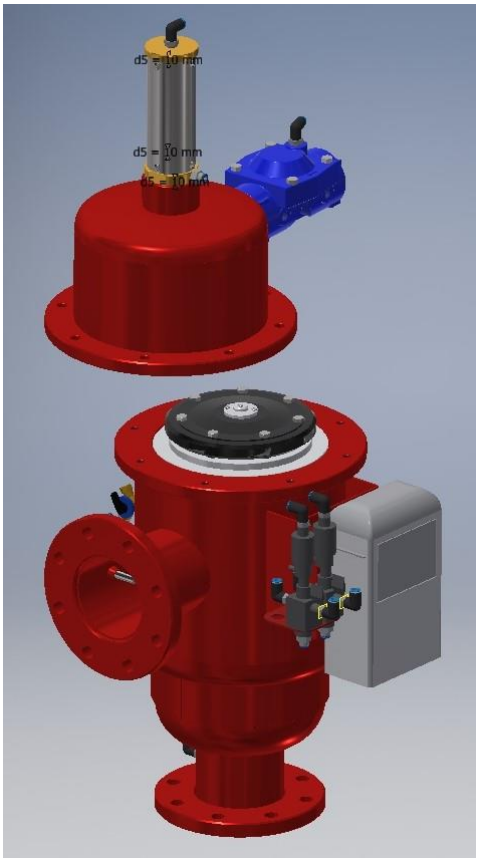
Before any maintenance work, ensure that the filter is not working and before any disassembly ensure that the pressure inside has relieved.

- Close the valve inlet and outlet valve
- Regularly inspect the finger filter at inlet and if necessary clean it. The finger filter serves to prevent dirt from entering the differential pressure switch which will result in improper operation of the controller.

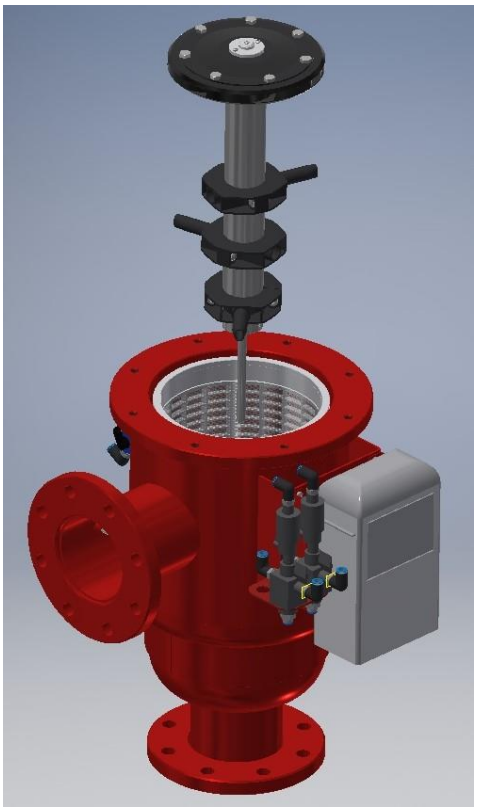
If it is necessary to dismantle the filter and to clean by hand the stainless steel cartridge, follow the sequence described below. Reassemble in reverse order.



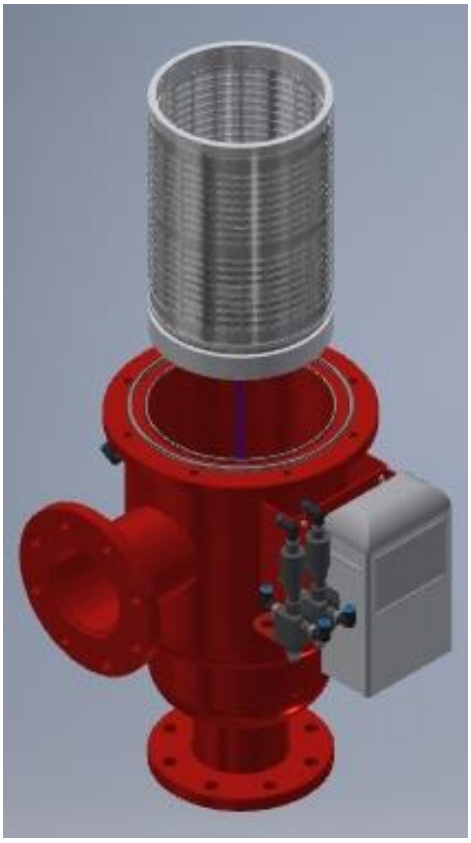
Unscrew the bolts and washers (8 X)



Remove the rotor chamber and the hydraulic cylinder



Remove the rotor cleaning assembly



Remove finally the cartridge

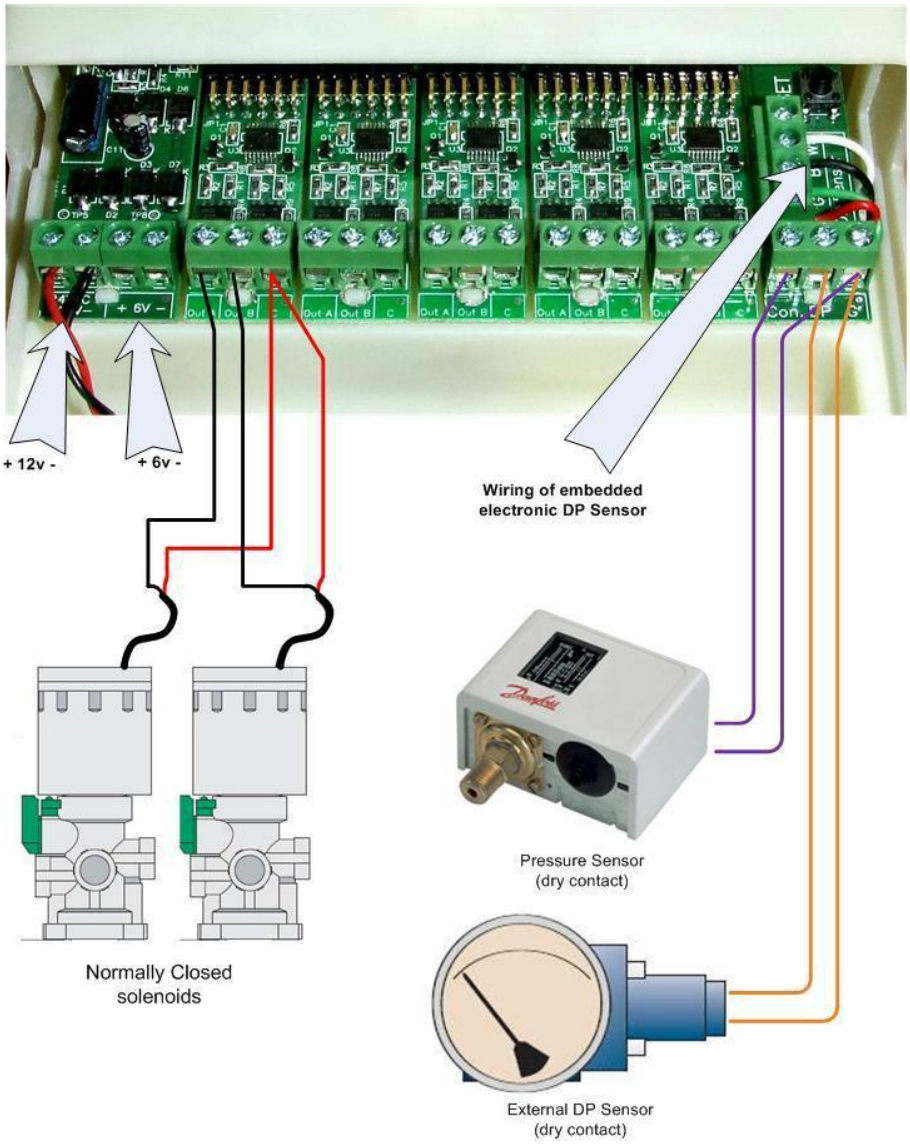
WIRING DIAGRAM

DC MODEL

The drawing below shows the wiring of the DC model of the controller.

Notice that:

1. The External DP sensor is optional and it is intended for use in cases there is no Embedded Electronic DP included.
2. The powering of the unit can be either by 6v DC or 12v DC.
3. The solenoids will be of 12v DC latch

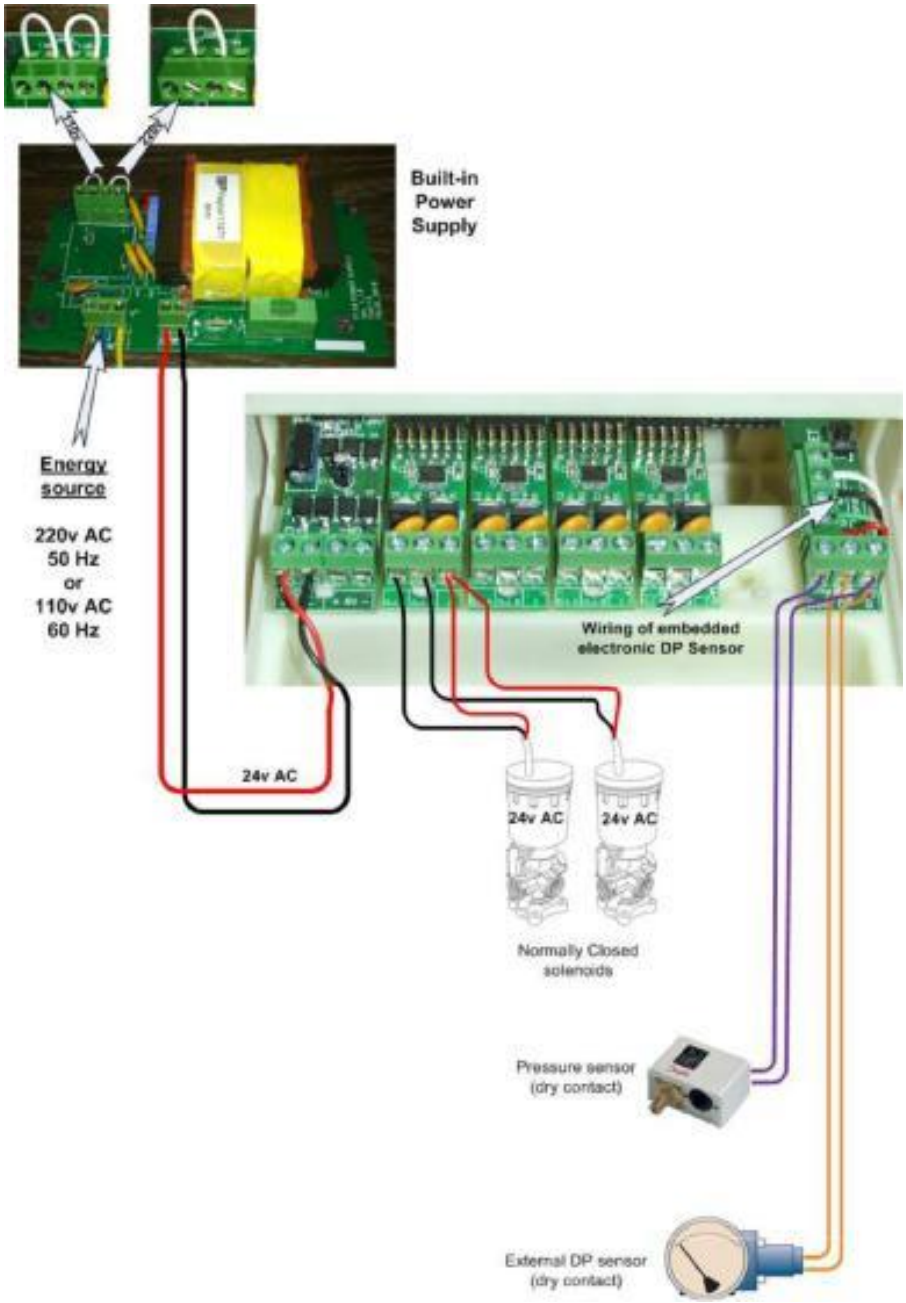


AC MODEL

The drawing below shows the wiring of the AC model of the controller.

Notice that:

1. The External DP sensor is optional and it is intended for use in cases there is no Embedded Electronic DP included.
2. The powering of the unit is by 24v AC transformed from 220/110 v AC.
3. The solenoids will be of 24v AC.



HYDRAULIC CONNECTIONS

